



OXFORD JOURNALS
OXFORD UNIVERSITY PRESS

RANKING THE SCHOOLS: HOW SCHOOL-QUALITY INFORMATION AFFECTS SCHOOL CHOICE IN THE NETHERLANDS

Author(s): Pierre Koning and Karen van der Wiel

Source: *Journal of the European Economic Association*, April 2013, Vol. 11, No. 2 (April 2013), pp. 466-493

Published by: Oxford University Press

Stable URL: <http://www.jstor.com/stable/24538800>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

Oxford University Press is collaborating with JSTOR to digitize, preserve and extend access to *Journal of the European Economic Association*

RANKING THE SCHOOLS: HOW SCHOOL-QUALITY INFORMATION AFFECTS SCHOOL CHOICE IN THE NETHERLANDS

Pierre Koning
VU University Amsterdam

Karen van der Wiel
CPB Netherlands Bureau for Economic
Policy Analysis

Abstract

This paper analyzes whether information about the quality of high schools published in a national newspaper affects school choice in the Netherlands. We find that negative (positive) school-quality scores decrease (increase) the number of first-year students who choose a school after the year of publication. These effects are only large for the college-preparatory track, such that a school receiving the most positive score for its most academic track sees 16–18 more first-year students enroll. We find that parents respond to the most recent and most prominently displayed information. The effects of information about school quality do not seem to be greater in regions with larger relevant newspaper circulation, suggesting that direct exposure to news about school quality does not explain the response to this information. (JEL: I20, D10, D83)

1. Introduction

Information about the quality of public institutions such as hospitals and schools has become increasingly widespread. Many observers expect this trend to improve the overall quality of public services, as it is assumed that with better information managers of public institutions are better able to benchmark their performance to that of their peer institutions, taxpayers can better hold those managers accountable, and, last but not least, consumers may make better-informed choices. As the supply of quality information has grown, so has literature concerning the effects of this information on consumer choice (e.g., Kling et al. 2008; Jin and Leslie 2003, 2009; Pope 2009). Regarding school choice, indirect effects involving house prices are relatively well

The editor in charge of this paper was Stefano DellaVigna.

Acknowledgments: This paper was written with the financial support of the Dutch Ministry of Economic Affairs. The authors thank those who provided the data for the analysis: Statistics Netherlands, The Dutch Inspectorate of Education, and the newspaper *Trouw*. Conversations with the Inspectorate, *Trouw*, *Elsevier*, and Jaap Dronkers were very useful. Ali Aouragh and Jeannette Verbruggen provided excellent research assistance. The authors would also like to thank Arthur van Soest, Casper van Ewijk, Dinand Webbink, Marc van der Steeg, Adam Elbourne, Maarten Cornet, André de Moor, Inge Groot, Broos Brouwers, Floor Westendorp, Joost Baeten, and seminar participants at CPB, Utrecht University, and Tilburg University for useful comments and suggestions. The paper benefited greatly from the comments of the editor and the three anonymous referees. All remaining errors are our own. Koning and van der Wiel are Research Fellows at IZA.

E-mail: P.W.C.Koning@vu.nl (Koning); K.M.van.der.Wiel@cpb.nl (van der Wiel)

Journal of the European Economic Association April 2013 11(2):466–493
© 2013 by the European Economic Association

DOI: 10.1111/jeea.12005

established (e.g., Downes and Zabel 2002; Figlio and Lucas 2004; Kane, Riegg, and Staiger 2006). More recently, some studies have investigated the direct effects of school-quality information on school-choice behavior, with Hastings, Van Weelden, and Weinstein (2007) and Hastings and Weinstein (2008) as important contributions. Both studies analyze an experiment in which parents of students at low-performing schools are provided directly with information about the quality of alternative schools. They find that the provision of this explicit information led more parents to choose higher-scoring schools for their children, by five to seven percentage points.

Like Hastings and co-authors, we study the direct effect of school-quality information on school choice, using a longitudinal dataset comprising first-year student inflow into all secondary schools in the Netherlands.¹ In doing so, we make three contributions to the literature regarding information and consumer choice. First, with an average of ten relevant schools within a radius of ten kilometers, the Netherlands presents an interesting setting to study the direct effects of school-quality information on school choice. Negligible school fees, good public transport, and free school choice ensure that parents are able to respond to information about school quality. This contrasts to the US or UK setting, where high-quality schools are often oversubscribed and where parents may play strategic games with other parents to improve their match. At the same time, an analysis based on the Program for International Student Assessment (PISA) sample shows that, compared with other OECD countries, the variation in student attainment between schools in the Netherlands is considerable and persistent (Field, Kuczera, and Point 2007).² Dutch parents thus have the opportunity to choose between schools and their choice matters for the quality of education enjoyed by their children.³

The second contribution of this paper is that we address the role of the media in the dissemination of school-quality information. In doing so, we follow up on recent empirical papers regarding the economics of the media (e.g., Gentzkow 2006; DellaVigna and Kaplan 2007; Enikolopov, Petrova, and Zhuravskaya 2011). The quality information under review in this paper is published by a small, national newspaper named *Trouw*. In 2009, *Trouw*'s market share amounted to 2.5% of all paid newspaper copies. This means that the published school rankings are not actively delivered to most parents' doorsteps. However, *Trouw*'s special issue on 'school

1. Koning and Van der Wiel (2012) use this same dataset to analyze how school boards respond to these school-quality rankings in terms of their subsequent performance in later school-quality rankings.

2. The fact that twelve-year-old students sort into different school tracks, a phenomenon present in other European countries such as Germany and Finland, and that many schools only offer a single track, may well exaggerate the objective quality differences between schools. Differences between identical tracks at different schools would be considerably smaller.

3. One could argue that quality differences should not persist in equilibrium in a country in which school choice is free. This is probably true if all players in the market have information about school quality. However, *Trouw* only started publishing objective school-quality indicators in 1997. It may take time for the educational market to adjust, as information trickles down slowly and school capacity cannot be enlarged overnight. Recent oversubscriptions in excellent academic tracks suggest that there is indeed long-term development in the market towards higher academic-quality elasticity of demand (see Section 2 for a more detailed discussion).

performance' is relatively well known among parents of children about to go to high school, indicated by—and perhaps because of—the degree to which regional newspapers copy the information into their own pages. Differences in this regional media coverage are exploited in this paper to gain more insight into how information spreads through society.

Finally, we contribute to the discussion regarding the effects of information types on school choice and we also consider the ex-post rationality of these choices. We can examine which of two summary statistics—a gross quality score or a prominently displayed final score—generates the largest response in terms of new students. The data also allow us to analyze whether parents pay attention to the most recent publication or to the most recent, yet unpublished information. This is because *Trouw*'s school ranking is based on quality information that is two years old, which displays substantial volatility from one year to the next. Recent literature has focused on this volatility in ratings (e.g., Chay, McEwan, and Urquiola 2005). Using our data, we assess the predictive value of the school rankings for future educational quality, testing whether the observed responses to the information were rational ex post.

Trouw uses objective quality indicators from the Dutch Inspectorate for Education to calculate an overall quality score for every track offered at each school. These scores are likely to be correlated with other, unobserved characteristics of the track, such as its reputation or the quality of its management. We thus potentially overestimate the effect of the *Trouw* scores on student inflow. By estimating fixed effects of school tracks, we can control for any time-invariant omitted variables. Furthermore, the two-year lag between the registration of quality indicators and their publication in *Trouw* reduces potential overestimation due to time-varying omitted variables. Various robustness checks are performed to test for remaining bias in our estimates. We find that *Trouw*'s quality scores have a robust and significant effect on the number of first-year students. The analysis reveals that parents pay the most attention to the most prominently displayed and most frequently discussed summary statistic. Moreover, the most recently published quality information matters the most for school choice. Overall, we find information effects that are smaller than those obtained by Hastings and Weinstein (2008). This is to be expected, because we examine the effect of information published in the media, whereas Hastings and Weinstein analyze information that was directly and actively provided to parents.

The effect of information is strongest for the college-preparatory track ("VWO"), the one that prepares students for university. We will refer to this track as the most academic track, a name that remains closer to the Dutch title. When *Trouw* qualifies an academic track as "++" (i.e., the highest ranking), the inflow of students increases by 16–18 students in the year after publication. This increase is nonnegligible, given that the average class in our sample consists of 76 students. Two complementary explanations can be given for the large effect on the academic track. First, parents of smart and ambitious students care most about academic quality and are best at searching for information. This is the argument of Hastings, Kane, and Staiger (2005), who showed that the preference attached to schools' mean test scores increases with neighborhood income and with the student's own academic ability. Second, it could

be that parents are aware of the noisiness of the quality rankings that are relevant to them. Interestingly, the academic track is the only track for which *Trouw*'s scores are clearly representative of the long-term educational quality provided. For the other school tracks, there is only a limited relationship between their quality scores at time t and their subsequent quality scores. Hence, it might be rational for students in those tracks to pay less attention to the information published in *Trouw*.

We have estimated additional regressions that exploit natural variation in the relevant newspaper circulation in the Netherlands. Our estimates show that the response to the information in terms of new students is not actually larger in periods during which *Trouw* readership is higher or in periods during which larger newspapers in the region publish *Trouw*'s rankings. Direct exposure to the school-quality information does not seem to be the main mechanism behind its effective use. Instead, parents' active search for information and/or information disclosure by schools probably explains the effect of quality scores on student inflow. Unfortunately, we are unable to identify the relative importance of these demand- and supply-side factors. Still, we do find some evidence suggesting that the benefits of objective school-quality information are larger in more competitive markets.

Finally, we estimate conditional logit regressions on a cross-section of individual school choices to assess the relative importance of quality information vis-à-vis other school characteristics. We find that distance is by far the most important determinant of school choice and that the willingness to travel to well-performing schools is small. It is estimated that students are willing to travel about 220 meters farther in order to attend a well-performing rather than an average school.

The remainder of this paper proceeds as follows. Section 2 details the Dutch institutional environment and *Trouw*'s school quality scores. Section 3 explains which school-level data and which empirical design are used in the empirical analysis. The empirical results of school-level data are presented in Section 4 of the paper. Additional results regarding individual-level data are presented in Section 5, while Section 6 concludes.

2. School Choice and Transparency of School Quality in the Netherlands

2.1. School Choice

The Dutch Constitution has guaranteed freedom of education since 1848. Initially, freedom meant that every group of citizens was allowed to establish a school following their own religious, societal, or educational beliefs. Since 1917, the Dutch state has even financed these "private" schools in the same way that it finances public schools that do not have a specific affiliation. The Netherlands also has a long history of free school choice. Students may freely decide which primary, secondary, and tertiary educational outlets they wish to attend.

Free school choice in the Netherlands is often regarded as a rather unique phenomenon, because in practice there are few limitations preventing students from

choosing another school besides the one that is located most closely (e.g., Bishop 1998; Dijkstra, Dronkers, and Karsten 2004). School fees are negligible for both public and private schools, so financial constraints are not binding. On average, children may choose between ten secondary schools offering the relevant track within a radius of ten kilometers from their home address. The absence of catchment areas and the large choice set render it unlikely that parents change residence to live close to their preferred schools or that the proximity of good schools is to any large extent reflected in higher housing prices.

It rarely happens that students are declined access to their most-preferred school; selection is only possible in the rare event that a school receives substantial oversubscription or when parents' beliefs evidently deviate from those of the school. Only in recent years has oversubscription become a problem for a select group of secondary schools that offer the most academic track in densely populated cities, such as Amsterdam and The Hague. We estimate that only 1% of schools in our sample must accommodate oversubscription. The popular press argues that the public dissemination of objective information about school quality has contributed to these waiting lists (Weeda 2010). After searching the LexisNexis database of national and regional newspapers, we find that the timing of relevant events lends some credence to such a hypothesis: the first *Trouw* publication regarding school quality was released in 1997, national newspapers first mentioned oversubscription of schools in 2000, and in 2002 parents brought the first lawsuits against oversubscription. Because we lack reliable data, our empirical analysis pays no further attention to oversubscription.

In the Netherlands, the choice of which high school to attend is made simultaneously by the parents of all twelve-year-old children.⁴ Children in the final year of primary education have to wait for their "school advice" in order to enroll at a secondary school. This advice is compiled by their primary school teacher, who relies on the child's test scores on a centralized exam taken halfway through the year. Each student's school advice states which school track the teacher believes the student is able to complete. A secondary school typically requires its prospective students to have a school advice coinciding with the track it offers. Over the years, the categorization of school tracks has changed, but three broad categories can be distinguished that have been constant over time. The most academically oriented school track (in Dutch: "VWO"), from which a diploma guarantees admission to universities, lasts six years. The middle level, general school track (in Dutch: "HAVO"), which guarantees admission to a "hogeschool" (comparable to colleges), lasts five years. The lowest track that provides for a general education (in Dutch: "MAVO", and later on "VMBO-gt") lasts four years and prepares students for vocational tertiary education. We limit our analysis to these ordinally classifiable tracks.⁵ Note that in the first and second years, the majority of secondary schools offer multitrack classes.

4. We understand that the choice for a secondary school is typically influenced by both parents and children. For expositional reasons, we only mention parents as the decision makers in this paper.

5. A fourth track that provides vocational training exists. We have excluded this from our analysis for multiple reasons. Many policy changes have recently affected this track. The educational emphasis of this track is on skills, not on academic achievements. Also, quality differences are much harder to measure, as

2.2. School-Quality Information

Since 1997, the national newspaper *Trouw* has published information about the quality of each school track offered at all secondary schools in the Netherlands. *Trouw* did so after it won a court case against the Inspectorate of Education over the disclosure of its quality reports. Each fall, *Trouw* publishes a list of schools stratified by province in a separate section of its weekend edition. Note that in the Netherlands a large share of circulation is through long-term subscriptions. In 2009, 82% of *Trouw*'s copies were intended for subscribers. Although the newspaper does perform its own calculations, the publication is based on the school records of the Inspectorate of Education.⁶ All three tracks feature separately in the publication, so that a school offering all three tracks has entries in three different locations with potentially different quality scores.

Although the exact information printed by *Trouw* has changed from year to year, some variables have been recurring items for all years. First, *Trouw* has always printed background characteristics such as school size, religious affiliation, and the percentage of children from cultural minorities. Second, three quality indicators are observed and used for all years. These are the average grade students achieve on the centralized exam in their final year of education; the percentage of students who, from the third year onwards, leave the school on schedule and with a diploma; and the net percentage of students who in the third year are in school tracks that are above or below their school advice. The registration of the last two indicators prevents schools from gaming their average grade results by either excluding low-performing students from final exams or by forcing students into lower school tracks. Figure A.1, in the Online Appendix to this paper, shows an example excerpt of the *Trouw* publication from 2002.

Trouw calculates two overall quality scores by school track on the basis of these three objective performance indicators.⁷ First, a "gross" overall quality score is determined using factor analysis.⁸ According to this estimate, all school tracks are then distributed into five categories (ordered by improving quality: "–", "–", "0", "+", or "++"). Second, in order to provide a quality measure that is closer to the 'value added' by a school, the overall raw quality score is corrected for several factors correlated with the initial quality of students. This has typically been done through OLS regressions using the percentage of children from cultural minorities as a control variable.⁹ In a similar fashion as for the unadjusted scores, five final quality scores

large differences exist between the curricula of various schools (think of car mechanic theory versus sports teacher training).

6. More information regarding the quality information that the Inspectorate registers and the information that *Trouw* publishes can be found in Dijkstra et al. (2001). Dronkers (1999) performed an initial assessment of the association between the *Trouw* scores and student inflow.

7. Koning and Van der Wiel (2012) explain in more detail the estimation procedures and how they have changed over time.

8. Although the three performance indicators mentioned have always been included in the overall scores, other variables, such as the percentage of tardy students, were also included in several other years.

9. The percentage of children from cultural minorities has always entered the correction equation, but its definition has changed several times. Other controls that have been included in certain years are the percentage of students from low-income households and the students' school advice.

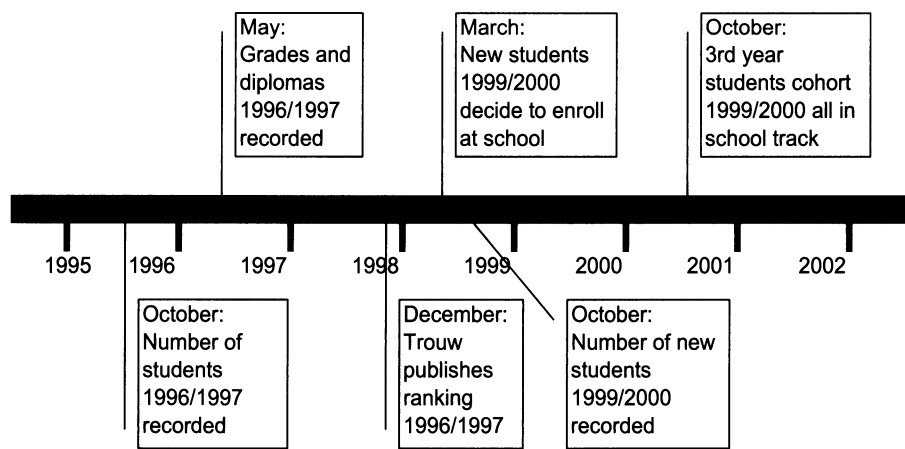


FIGURE 1. Time line for the publication of public school-quality information.

are assigned, ranging from “– –” to “++”. *Trouw* prominently presents the adjusted scores as the final quality scores. Unfortunately, we do not know the actual values of the variables underlying either the adjusted or the unadjusted scores. *Trouw* did not print these latent variables in its publication, did not store them, and has not documented the exact procedures they used. The procedure has also changed several times over our sample period. We have tried to reproduce the variables, but we failed to do so reliably.

There is a relatively long delay between the Inspectorate of Education’s registration of the quality indicators and *Trouw*’s publication of the quality scores. Figure 1 presents a time line of the relevant recurring events. It shows a three-year lag between the registration of data and the registration of the potential response to that information in terms of the number of new students. The Inspectorate takes about a year and a half to generate the school-quality records, *Trouw* spends another six months finalizing its publication, and students are only observed entering school ten months after that.

Although *Trouw* was the first media outlet to publish these quality rankings, there are two other objective sources of information parents could use. Following a change in policy, the Inspectorate of Education began publishing their own data on their website in 2000. This means that school-quality information can be separately reviewed for each school and each school track. However, the way the information is presented—the website shows many variables, and it is unclear what they represent—makes it hard to compare quality across schools, especially because an overall measure of quality was absent until 2006. Besides this, the national weekly magazine *Elsevier* started publishing rankings in 2001 based on the same information from the Inspectorate that *Trouw* uses. A major difference between the two publications is that, rather than measuring single years, *Elsevier* takes as inputs three-year moving averages of the quality indicators. Moreover, the final quality scores in the *Elsevier* rankings are unadjusted for the percentage of children from cultural minorities.

To assess the relative importance of the information published in *Trouw* versus that published in *Elsevier*, we have combined circulation data with information concerning

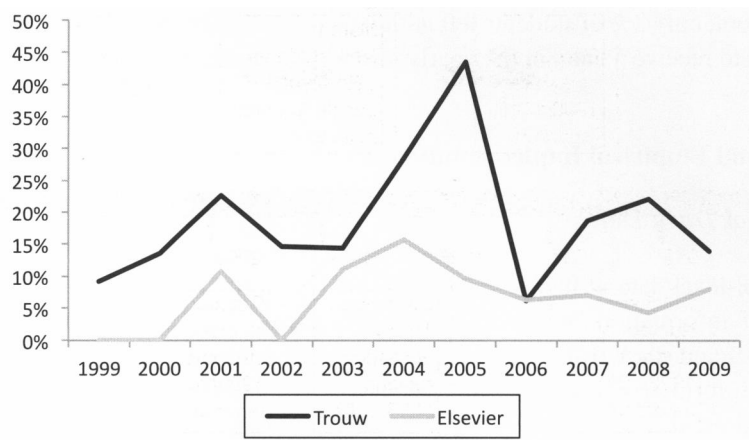


FIGURE 2. Circulation of *Trouw* and *Elsevier* school rankings, including all newspapers mentioning these publications, from 1999 to 2009, relative to total newspaper circulation in The Netherlands.

the content of all national and regional newspapers in the Netherlands (see Section 3 for a more elaborate description of this dataset). Comparing average direct circulation in 2005, *Elsevier* leads with 141,000 weekly copies compared with 121,000 daily copies of *Trouw*.¹⁰ A different picture emerges, however, if we take into account other media outlets that report on the school rankings. We tracked articles published between 1997 and 2009 and in other media outlets that mention the *Trouw* and *Elsevier* publications regarding school quality. For *Trouw*, this resulted in a total of 121 articles,¹¹ whereas we found 43 articles for *Elsevier*. Figure 2 represents aggregate circulation data of these newspapers relative to total yearly newspaper circulation in the Netherlands. It turns out that the total circulation of newspapers mentioning *Trouw*’s publication of school rankings is substantially larger than of those mentioning *Elsevier*’s, with a maximum of 2.4 million newspaper copies mentioning the *Trouw* rankings in 2005 (44% of all issues).¹² Besides the longer time period covered by the *Trouw* dataset of rankings, the newspaper’s larger circulation gives us another argument for focusing on the *Trouw* rankings. In Section 4, however, we will show some robustness checks that take into account any effects of the *Elsevier* rankings. Finally, it is important to note that regional newspapers are not the only entities copying and disseminating *Trouw*’s information. Anecdotal evidence suggests that schools and parent interest groups are also active in this regard. For example, one school website prominently states, “[. . .]

10. Interestingly, the ‘school performance’ publication has moved to a much larger newspaper, the *Volkskrant*, in 2012.

11. When reading these press clippings, it was striking to see how many articles (73%) highlight schools with high rankings and how few highlight schools with low rankings. Typically, regional newspapers present a list of local schools that have received a “+” or “++” ranking. Only in some cases is the distinction between these two rankings mentioned explicitly.

12. Also note that the actual number of newspapers mentioning the *Trouw* publication is higher, as local, free and weekly newspapers and magazines are not counted.

In our school, only 3% of students left without a diploma. Our school is the only one in the region to receive 5 stars in the yearly school performance test of Trouw.[. . .]”¹³

3. Data and Empirical Implementation

3.1. School Track Data

The school-level dataset that we use in our analysis was compiled by joining several sources of information. We received “quality cards” for the years 1995–2008 for each school and track from the Inspectorate of Education. For each year, these cards provided information about the geographic coordinates of a school, its religious denomination, the number of students, the percentage of students in each track, the percentage of students from cultural minorities, and each school track’s quality indicators. As explained earlier, these objective quality indicators served as inputs for *Trouw*’s overall quality scores. From *Trouw* we received a paper copy of each of their yearly publications of school rankings, from 1997 to 2008. As *Trouw* had not stored these scores electronically, we manually added the final adjusted scores and the gross unadjusted scores to our dataset.¹⁴ For each municipality, we added information from Statistics Netherlands about population size in the relevant age categories.

Our final dataset contains 9,542 yearly observations of schools recorded from 1996 to 2005 (but published with a delay of two years) and 12,339 yearly observations of tracks offered at these schools. About half the schools in our sample offer all three tracks, while 25% of the schools offer only a single track. Unfortunately, we cannot use data from 1995, because at that time all information was recorded at the school rather than at the track level. Furthermore, in our empirical analysis, we cannot use the latter three years of data, because, as described previously, there is a three-year lag between the recording of the objective quality measures and the recording of the enrollment of new students who could conceivably have responded to the publication of this quality information.

Table 1 presents descriptive statistics of the school-level dataset split into groups based on the five different final quality scores. *Trouw* divided all school tracks into five categories, each of distinctly different size. About 1% of observations were filed in the very worst category (“--”), with another 1% in the very best category (“++”), about 50% of average performance (“0”), and the remaining tracks split equally between the badly (“-”) and well-performing groups (“+”). As the classification was performed separately for each school track, the differences in the distribution of scores between the tracks are negligible. The sample includes a substantial number of observations for which the *Trouw* score is missing. These are mostly schools where only a few students took their final exams. *Trouw* decided not to report a final score for the smallest

13. Website for the Stedelijk Gymnasium Leiden in June 2011 (<http://www.gymnasiumleiden.nl/>).

14. Using *Trouw*’s final scores was necessary because *Trouw* has used more detailed information from the Inspectorate to compute the scores than that to which we had access.

TABLE 1. Descriptive statistics of school-track dataset sorted by *Trouw* score (1996–2005).

	Missing	Most negative	Negative	Neutral	Positive	Most positive
Observations	1,560	171	2,018	6,215	2,287	88
Academic track (“VWO”)	11%	1%	16%	51%	20%	1%
Middle track (“HAVO”)	14%	2%	17%	48%	18%	1%
Lowest track (“VMBO-gt”)	12%	1%	16%	52%	18%	1%
Number of first-year students						
Mean	51	56	72	82	83	75
St. dev.	37	31	36	38	40	75
Change in first-year students						
Mean	1.5	−3.0	0.3	1.0	1.4	6.9
St. dev.	15	16	17	18	19	25
Absolute change in first-year students						
Mean	10.5	12.0	12.5	13.4	13.6	13.3
St. dev.	12	11	11	13	13	23
Grade obtained on exams						
Mean	6.3	6.0	6.2	6.4	6.6	6.7
St. dev.	0.4	0.2	0.2	0.2	0.2	0.3
Diploma without delay						
Mean	71%	49%	60%	70%	77%	82%
St. dev.	18	17	16	15	14	16
Transition probabilities						
From Missing	0.59	0.01	0.09	0.22	0.09	0.00
From Most negative	0.09	0.09	0.41	0.38	0.03	0.00
From Negative	0.07	0.03	0.33	0.50	0.07	0.00
From Neutral	0.05	0.01	0.16	0.61	0.17	0.00
From Positive	0.06	0.00	0.07	0.50	0.36	0.02
From Most positive	0.11	0.00	0.03	0.20	0.57	0.09

schools, as the confidence intervals for grade and diploma results at these schools are considered too large. This is mirrored by the high standard deviation in average grades for this subgroup. The disclosure of quality information does not seem to depend on the level of quality, as the underlying quality scores do not differ substantially between the missing and neutral categories. A selection regression analyzing whether schools that received worse grades one year were more likely the following year to go missing from the dataset confirms this finding.

On average, 76 students entered each school track every year. Schools that received a negative score had smaller entering classes, on average, than other schools. Table 1 also displays information about the average changes in student numbers by quality-score category. The unconditional changes in student inflow are quite similar to the fixed-effects estimates of quality scores obtained in Table 2. Tracks in the most negative

category lost students, whereas tracks in the most positive category gained students. Looking at absolute numbers, the class of first-year students was on average 13 students bigger or smaller than in the previous year for the best- and worst-performing schools, respectively. This is a non-negligible number, considering the average class size of 76. For the school tracks that received a final quality score, the average grades and the percentage of students that received a diploma without delay increased monotonically with the score ranking, as expected.

Table 1 also shows the transition probabilities that a track might change from one score one year to another score the following year. There was substantial variation in score ratings for each school track over time. In particular, the probability that a school received the same ranking the next year ranges from 9% (for the most negative and most positive categories) to 61% (for the neutral category). This raises questions about the noisiness of the *Trouw* rankings.¹⁵ Still, although the exact *Trouw* location in the quality distribution is largely transitory, there were only a few jumps from positive to negative scores and vice versa. The probability of moving from any positive to any negative score, or the reverse, was smaller than 7%.

To get a clearer picture of the persistency of *Trouw* scores, Figure 3 presents average *Trouw* scores over time for each school track separately. For the group of school tracks that received one out of five *Trouw* scores at time t , we computed the average numeric school quality in years following t (where “--” = 1, “-” = 2, “0” = 3, “+” = 4, and “++” = 5). If the *Trouw* scores do not yield any information about future quality, the lines representing different (initial) quality scores at time t should converge towards each other. Although there is some convergence in subsequent years to the average score (“0” = 3), quality differences do persist over time. For academic tracks, this trend held up until the end of the sample, whereas for the other two school track categories, differences in average future quality faded out by around period $t + 5$. Table A.1 in the online appendix provides an additional, more detailed analysis regarding the proportion of tracks obtaining a “++” score in each subsequent period that is relevant to students who choose a school track using information obtained at period t . There are six relevant time periods for students choosing the academic track, because it takes six years to complete this track, while it takes, respectively, five and four years for the other tracks. The same picture emerges here. Only for the academic track is the proportion of schools receiving a “++” score in future periods significantly higher for schools that scored a “++” in period t compared to schools that scored a “+”.

After the school-level information, we are interested in the variation in exposure to the *Trouw* publication between regions and over time. For each province and year we have constructed newspaper coverage data from the HOI (*Institute for Media Auditing*) and Cebuco (*Organization for the marketing of Dutch daily newspapers*) databases. We use two province-level variables for news coverage of school rankings; these are

15. See also Figlio and Lucas (2004), who argue that the large stochastic component of schools’ test grades limits the effects of school quality on house prices. We discuss this issue in further detail in Sections 4 and 6.

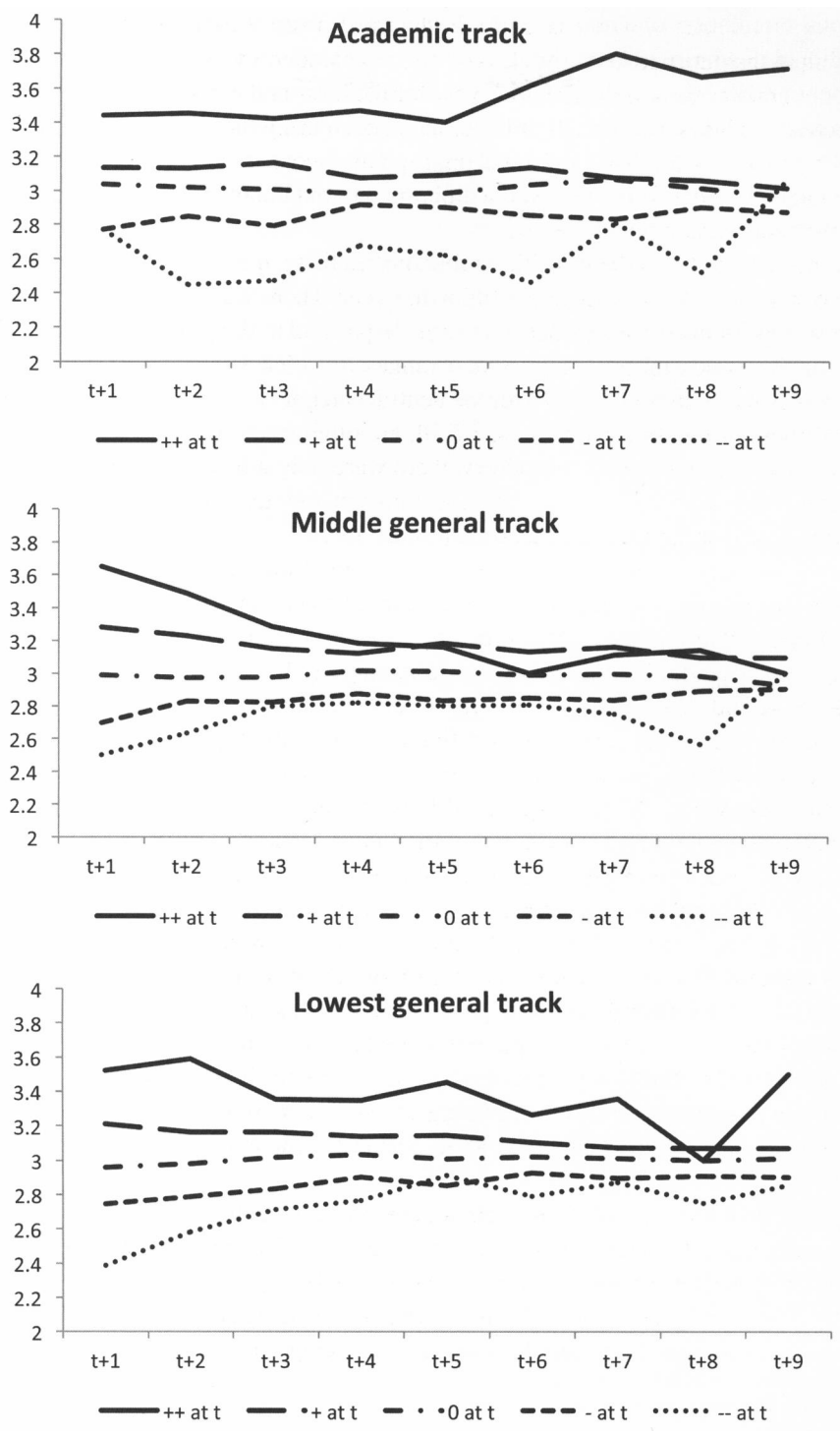


FIGURE 3. Average cardinal *Trouw* scores in subsequent years (“--” = 1, “-” = 2, . . . , “++” = 5).

the total circulation of *Trouw* per inhabitant older than 13, and the total circulation of regional newspapers mentioning *Trouw* per inhabitant older than 13. To assess the number of other newspapers copying *Trouw*'s school-quality information, we searched the LexisNexis database for all articles that mentioned *Trouw Schoolprestaties*¹⁶ and were published within two weeks of *Trouw*'s publication. Table A.2 in the Online Appendix summarizes the circulation and population data for each province. It shows that there was substantial variation in the relevant regional coverage between provinces and over time. One can easily imagine that the intensity of media coverage of *Trouw*'s ranking depends on the quality of schools in the region. We have therefore tested for an effect on circulation of a region's average score and of a region having many schools in the most negative or most positive categories; see Table A.3 in the Online Appendix for these results. This test only yielded significantly positive outcomes for the most positive score. That is, newspapers in regions with a high fraction of schools in the most positive category were more likely to report those outcomes.

3.2. Empirical Implementation

This section explains how we identify an effect of school-quality scores on school choice in the year after *Trouw*'s publication. We estimate *school-track*, fixed-effect regressions of the number of first-year students in each track.¹⁷ The number of first-year students y for schools i , school tracks j , and time periods t serves as our dependent variable:

$$y_{ijt} = \sum_{r=-}^{++} \alpha' R_{r,ij,t-3} + \chi' NA_{ij,t-3} + \lambda' x_{ijt} + \kappa' x_{ij,t-3} + \sum_{t=1996}^{2005} \gamma' T_t + v_{ij} + \varepsilon_{ijt}. \quad (1)$$

In equation (1), all R_r are dummy variables representing the occurrence of quality scores r per school track j . The dummy "NA" equals one if no quality score is provided for school track j at school i . The x -variables represent time-varying controls either from this period or from the period in which the quality information was recorded. The x -variables include market proxies, such as the size of the adolescent population in the municipality, the number of schools within a 20 km radius of the school at time t , and the average *Trouw* quality score within that radius, as well as school characteristics at time $t - 3$, such as the total number of students that attended the school,¹⁸ the number

16. In English: "*Trouw* school performance".

17. Besides the school-track regressions, we have estimated school-level, fixed effect regressions of the total number of new, first-year students at each school. The dependent variables in those regressions were (among others) the quality scores received by each school track within that school. This procedure leaves room for spillover effects between one track's score and the inflow into other tracks. The results were not qualitatively different from the *school-track* regressions. For expositional reasons we have left these results out of the paper.

18. One may worry about the fact that the number of students at $t - 3$ is a function of lagged dependent variables, causing our within estimator to be inconsistent. We have estimated models with and without the

of school locations under management, the percentage of students in each track at the school, and the percentage of children from cultural minorities.¹⁹ We also include all yearly time dummy variables T . As we estimate school-track fixed effects, there are two separate error terms: the time-invariant, school-track-specific term v_{ij} and the error term ε_{ijt} , which we assume to be i.i.d. $(0, \sigma_\varepsilon^2)$. Furthermore, in the school-track regressions we correct our standard errors for clustering at the school level.

Since we cannot control for unobserved characteristics we might overestimate the effect of the quality scores on student numbers.²⁰ Using fixed effects enables us to control for any time-invariant, omitted variables, such as school reputation, the quality of school management, or the physical school environment, that could influence student inflow. Next, to correct potential biases due to time-variant unobserved factors, we exploit the three-year lag between the realization and publication of information about quality and the observed school choice. This long lag breaks down a substantial part of the instantaneous correlation between omitted, time-variant variables that influenced both student inflow and the quality score published at time t . Even so, we need to verify that there is no remaining serial correlation in the quality scores over time. Moreover, we need to test for the assumption that school-specific quality trends are uncorrelated with trends in the number of new students. In the empirical results section of this paper, we therefore present a variety of robustness checks. To start with, we include several placebo specifications using both the *Trouw* scores published at time t and the *Trouw* scores published at times $t - 1$, $t + 1$, and $t + 2$. These regressions test for any considerable serial correlation problem in the original estimates. We furthermore present estimates both of the effect of the unadjusted quality scores and of the effect of *Elsevier's* quality scores on student inflow. Finally, we include a specification with a school-specific time trend, one with a school-track-specific time trend, a first differences specification, and a log linear specification.²¹

In our analysis, we do not directly observe the number of first-year students in each school track, as many schools only offer mixed-track classes in the first year. Therefore, we proxy the number of first-year students within a track in academic year t by the number of third-year students in each track in period $t + 2$. For each school track, we thus lose two yearly observations. The number of first-year students is observed with a measurement error, as in two years' time the school-track cohort will have lost and/or added some students. We are able to test whether this measurement

number of students at $t - 3$ as a control variable. The quality-score estimates are somewhat stronger when excluding the number of students at $t - 3$.

19. It should be noted that the definition of cultural minority students changed in 2003 and 2005. The average value of this variable is therefore not presented in Table 1. In the estimation of our models, we also control for this variable by allowing its impact to vary from year to year.

20. Table A.4 in the Online Appendix column (III) indeed shows that the OLS estimates that correspond to those in our baseline fixed-effect regression in column (I) in Table 2 are typically larger than those estimates using fixed effects.

21. In order to take individual time trends into account, we estimate regressions of detrended and demeaned student inflow numbers on the detrended and demeaned independent variables. This method is equivalent to estimating separate school track time trends, and it is feasible for the large set of schools in our sample.

TABLE 2. Main model: regression coefficients from school-track fixed-effect regressions (dependent variable: first-year students in school track).

	(I)	(II)	(III)	(IV)
	All school tracks b/se	Most academic track (VWO) b/se	Middle academic track (HAVO) b/se	Lowest general track (VMBO) b/se
<i>Quality score at time t</i>				
Not available	−0.812 (0.89)	0.616 (1.55)	−2.034 (1.61)	−0.997 (1.07)
Most negative	−3.901* (1.65)	−6.188 (3.64)	−5.412 (3.08)	−1.353 (1.95)
Negative	−3.021*** (0.52)	−4.252*** (0.90)	−2.927** (0.93)	−2.293** (0.79)
Neutral	Reference	Reference	Reference	Reference
Positive	1.903*** (0.52)	3.096*** (0.90)	0.549 (1.02)	1.829* (0.74)
Most positive	6.555* (2.67)	17.730*** (5.32)	5.137 (5.32)	1.578 (2.27)
<i>F</i> -stat quality scores	11.28	9.12	2.59	3.35
<i>p</i> -value	0.00	0.00	0.02	0.01
<i>N</i>	12,339	3,648	3,754	4,937

Notes: Additional controls include the size of the adolescent population in the municipality at t , the number of schools within 20 km at t , the average school quality within 20 km at t , the total number of students that attended the school at $t - 3$, the number of school locations the school management operated at $t - 3$, the percentage of students in each school track at $t - 3$, the percentage of children from cultural minorities at $t - 3$, and all time dummy variables.

Standard errors between parentheses, clustered at the school level.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

error is independent of the other variables in the regression, since for the subsample of single-track schools we observe both the proxy and the true value of student numbers in a track. From regressions of the observed squared errors, we conclude that quality scores are not correlated with the measurement error in student numbers.²² This means that the measurement error only affects the efficiency, and not the consistency, of our estimates.

4. Empirical Results

4.1. Main Model

In this section, we investigate whether publicly available quality information has a causal effect on the number of students choosing any particular school in the Netherlands. The results of the main model are shown in Table 2. Column (I) presents estimates for all school tracks together, whereas the other three columns focus on one of the three particular school tracks. The first column shows that there is indeed a

22. The outcomes of this analysis are available upon request.

significant, albeit small, effect of the quality scores on the number of students that enroll in that particular track. The school track cohort of new, first-year students is estimated to be four students smaller when a track scores a double minus (“--”) than when a track receives a neutral score. A single minus (“-”) results in a new student cohort that is three students smaller. We find that the cohort of new students grows by two students when a school track scores a plus (“+”). These are small effects when compared to the average number of 76 first-year students attending a school track. The largest effect is found when *Trouw* qualifies a school track as excellent (“++”), with seven more students attending that particular school track in the year after *Trouw*’s publication. The joint effect of all five quality-score categories is clearly significant, as the *F*-test statistic indicates.

When evaluating the estimates for the separate school tracks, the response to information about quality for all scores is largest for the most academic school track (Column II). VWO-schools with a negative score lose four new students, on average, whereas VWO-schools with a positive score win three students. We furthermore estimate that eighteen more students choose a school in the most academic track in the year after *Trouw* has given it a “++” rating.²³ This is a substantial number, especially given that the recent emergence of oversubscription at certain VWO-schools may bias these results downwards. For this response to be rational, the probability of obtaining a “++” score in subsequent periods should be highest for the “++” group up until period $t + 8$ as well.²⁴ The length of time results from the combination of the delay in the publication of quality information and the time it takes to complete an academic track. Table A.1 in the Online Appendix already compared the ex-post probabilities of obtaining a “++” score for both good and excellent academic tracks. Although the majority of tracks that were initially rated as excellent do have their rankings fall in later years, we observe a significantly larger probability of later being rated excellent. The stronger persistency of quality differences for the academic track is also apparent in the average expected scores (see Figure 3). The large “++” information effect for academic tracks could thus be rational.

The information effects found in Table 2 are smaller for the middle academic and lowest general tracks: three and four students refrain from choosing a HAVO and VMBO school because of a negative score, respectively. Two additional students choose a VMBO-school after it has received a positive score in *Trouw*. However, even for these school tracks and their smaller effects, the *F*-test of the null hypothesis that the coefficients of the quality scores are jointly zero is rejected at the 5% level. Two main arguments could explain why the information effect is largest for the most academic track. On the one hand, the type of student attending such a track might respond more strongly to information about school quality, either because the parents

23. A larger effect is found when focusing on schools that only offer the most academic track. The response in terms of new-student numbers is minus eleven when such a school scores a minus and plus 22 when such a school scores a double plus.

24. The lower general track takes four years to complete, the middle general track takes five, and the most academic track takes six years.

TABLE 3. Robustness checks: Regression coefficients from school-track, fixed effect regressions (Dependent variable: first-year students in school track).

Additional score	(I) <i>Trouw at t – 1</i> b/se	(II) <i>Trouw at t + 1</i> b/se	(III) <i>Trouw at t + 2</i> b/se
<i>Quality score at time t</i>			
Not available	–0.892 (1.02)	–0.806 (0.90)	–0.951 (0.92)
Most negative	–3.243 (2.08)	–3.626* (1.71)	–4.085* (1.68)
Negative	–2.513*** (0.59)	–3.086*** (0.53)	–3.231*** (0.54)
Neutral	Reference	Reference	Reference
Positive	1.724** (0.60)	1.901*** (0.53)	1.892*** (0.53)
Most positive	5.897 (3.08)	6.361* (2.72)	6.419* (2.66)
<i>Lags and leads of quality score</i>			
Not available	0.09 (0.83)	–0.927 (0.93)	–1.703 (0.90)
Most negative	–1.713 (1.73)	0.963 (1.75)	–3.102* (1.53)
Negative	0.097 (0.09)	–1.727*** (0.927)	–1.562** (1.703)
Neutral	Reference	Reference	Reference
Positive	0.428 (0.50)	0.838 (0.52)	0.133 (0.53)
Most positive	0.439 (2.14)	0.047 (2.55)	–1.718 (1.82)
<i>F</i> -stat scores time <i>t</i>	7.15	11.47	11.73
<i>p</i> -value	0.00	0.00	0.06
<i>F</i> -stat lag and leads	0.38	3.59	2.81
<i>p</i> -value	0.86	0.00	0.02
<i>N</i>	10,526	12,339	12,339

Notes: Additional controls include all variables included as in Table 2.
Standard errors between parentheses, clustered at the school level.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

of these relatively bright and ambitious students attach a higher value to academic achievements or because they experience a lower cost of acquiring such information. This argument is often found in the literature, as in Hastings, Kane, and Staiger (2005). On the other hand, parents choosing between several middle or lower general tracks might be aware of the fact that the quality scores for these tracks are largely transitory. Figure 3 has demonstrated that *Trouw*’s quality scores for the middle and lower general tracks are not significantly different from each other in terms of how they predict school quality in the future.

To test whether the responses to information can be uniquely attributed to the *Trouw* publication in each relevant year, we have performed several regressions with placebo effects of leads and lags (see Table 3). Besides the effect of the quality score published in period *t*, the table presents the effect of quality scores that were published

in periods irrelevant to the inflow of students. School-track scores published in the year prior to the decision year (Column (I)) generate no significant effect on student inflow. School-track scores published in the first (Column (II)) and second (Column (III)) years after parents have made their choices generate mostly insignificant and much smaller effects on student inflow than those published right before parents make their decisions. The latter two sets of estimates especially strengthen the robustness of our conclusions, as there is likely to be substantial correlation between the data underlying the *Trouw* scores at $t + 1$ and $t + 2$ (and gathered at $t - 2$ and $t - 1$) and unobserved, time-variant factors determining school choice. Parents thus respond mostly to the most currently published quality information.

We report more evidence regarding the types of information parents actually pay attention to in Table 4. The estimates in Column (I) include those for a regression of new, first-year students on the quality scores published by *Elsevier* magazine. The response to *Elsevier*'s scores is smaller and insignificant for all parameters when comparing these estimates to the baseline results generated using *Trouw*'s scores (in Table 2). A similar picture emerges if we compare results estimated over identical time periods. *Trouw* scores thus seem to be the more important sources of information. *Trouw* scores are probably better known because they have been around for a longer period of time, or perhaps because the coverage of *Trouw* in other newspapers is larger (Figure 1). These outcomes could further suggest that parents respond more to recent, but noisy information (*Trouw* is based on the most-recent quality observations) than they do to dated, but probably more robust information (*Elsevier* is based on a moving average of the last three years of quality information).

Column (II) of Table 4 presents evidence regarding the importance of prominence of display. The estimates are for a regression of new, first-year students on both the final adjusted and the unadjusted quality scores that *Trouw* publishes. As mentioned earlier, in Section 2, the newspaper presents the final score, which is adjusted for the ethnic composition of the student population, more prominently (in a bold end-column) than the unadjusted score. Comparing the results of the main model with those in Column (II), one may conclude that the response to the unadjusted scores is smaller than that to the final scores. An F -test on the joint significance of the gross scores is rejected. This suggests that the final score is better known, or that parents more highly value the adjusted "added value" scores.

Finally, we performed several additional robustness checks, as presented in Table 5. Columns (I) and (II) show estimates from the fixed effects regressions that are extended with school and school-track time trends. These were obtained by first estimating constants and time trends for all dependent and independent variables for each school track and then estimating the main model on the detrended residuals. Although some of the coefficients are no longer significant, the results in both columns look qualitatively similar to those of our main model. Column (III) displays first-differences estimates. Comparing the outcomes of the fixed-effects specifications with the first-differences estimates of our model essentially provides a test of the biasing impact of serial correlation on our coefficient estimates. In general, the coefficients in

TABLE 4. Robustness checks: Regression coefficients from school-track fixed-effect regressions (dependent variable: first-year students in school track).

Quality score Specification	(I) <i>Elsevier at time t</i> Fixed effects newspaper b/se	(II) <i>Trouw at time t</i> Incl. final and unadjusted score b/se
<i>Quality score at time t</i>		
Not available	−0.140 (1.37)	−0.474 (1.11)
Most negative	−0.445 (1.12)	−3.155* (1.57)
Negative	0.335 (0.66)	−2.619*** (0.57)
Neutral	Reference	Reference
Positive	1.098 (0.70)	1.512** (0.55)
Most positive	1.144 (0.99)	5.838* (2.62)
<i>Unadjusted score at time t</i>		
Not available		−0.133 (1.18)
Most negative		−2.703 (1.97)
Negative		−0.568 (0.72)
Positive		0.847 (0.60)
Most positive		2.265* (1.06)
<i>F</i> -stat quality scores	0.76	6.60
<i>p</i> -value	0.58	0.00
<i>F</i> -stat adjusted scores		1.35
<i>p</i> -value		0.24
<i>N</i>	7,798	12,339

Notes: Additional controls include all variables included as in Table 2.
Standard errors between parentheses, clustered at the school level.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Column (III) are comparable to those of our main model, so serial correlation does not seem too problematic here.

Column (IV) presents coefficients estimated in a school-track, fixed effects regression of the log of the numbers of new, first-year students. This column therefore shows the effects of the quality scores on student numbers, which are proportional to the average size of the new cohort. When a track receives the most negative (or a negative) score, 8% (4%) fewer students choose to attend that track in the following year, while 3% more students attend a track after it has received the most positive score. The largest relative effect is again found when a school track is rated as excellent (“++”); the group of new, first-year students then grows by 9%.

The results in Tables 3–5 are consistent across the various specifications, lending credibility to the notion that the estimates in our main model reflect a true, causal

TABLE 5. Robustness checks: regression coefficients from regressions for all school tracks (dependent variable: first-year students in school track).

Specification	(I) School-track trend Fixed effects b/se	(II) School trend Fixed effects b/se	(III) First differences OLS b/se	(IV) Log of first-year students in track Fixed effects b/se
<i>Quality score at time t</i>				
Not available	−0.334 (0.56)	−0.553 (0.62)	0.344 (0.47)	−0.010 (0.01)
Most negative	−1.933 (1.10)	−1.049 (1.17)	−4.234** (1.47)	−0.081* (0.03)
Negative	−1.615*** (0.38)	−1.858*** (0.43)	−0.920* (0.46)	−0.042*** (0.01)
Neutral	Reference	Reference	Reference	Reference
Positive	0.721 (0.40)	0.997* (0.45)	0.444 (0.48)	0.027*** (0.01)
Most positive	5.459* (2.23)	4.138 (2.37)	6.054* (2.94)	0.092** (0.03)
<i>F</i> -stat scores	5.56	5.52	3.94	11.04
<i>p</i> -value <i>F</i> -test	0.00	0.00	0.00	0.00
<i>N</i>	12,339	12,339	10,437	12,339

Notes: Additional controls include all variables included as in Table 2.

Standard errors between parentheses, clustered at the school level.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

effect of information about quality, as opposed to an effect of unobserved school characteristics. All in all, the robustness checks confirm the general finding that, when choosing a school for their child, parents do pay attention to relevant information about school quality and, furthermore, that they respond to newspapers’ most recent and most

4.2. Mechanisms Driving the Effect of Information about Quality

The previous section shows that information about school quality has an effect on school choice in the period directly after the publication of the information. Others in the literature (e.g., Figlio and Lucas 2004; Hastings and Weinstein 2008) have documented such a direct effect of the transparency of information about quality on economic outcomes. In this subsection, we add more information regarding the mechanism behind the effect in our sample. In this respect, it is important to keep in mind that the *Trouw* information is not directly distributed to parents. Thus, the use of this information crucially depends on how it is disseminated among them.

To start with, one of the possible mechanisms for information dissemination is through direct exposure: it could be that parents choose an appropriate school after reading the relevant issue of *Trouw*. A way to test whether direct exposure is the

most important mechanism is to exploit variation in the relevant newspaper coverage in the Netherlands. There are two sources of variation that we can use for this test. First, the originally Protestant newspaper *Trouw* is widely read in certain areas of the country and less read in other, predominantly Catholic, areas. Its readership has also fallen over recent years. Our first test therefore uses the within-region, time-variant differences in the quantities of *Trouw* circulation. Second, many regional newspapers copy information from the *Trouw* publication when that information is relevant to their region and their readers. As shown in Sections 2 and 3, there is substantial variation over time and place in the number of newspapers mentioning the school-quality scores.

Table 6 presents interaction coefficients of the school-quality scores with three different normalized variables, such that the size of these coefficients can be interpreted as the effect of changing this variable by a standard deviation. Column (I) presents the estimation results of a regression including interactions of the *Trouw* scores with the *Trouw* coverage rate for each province (yearly *Trouw* circulation per inhabitant aged 13 and older). Column (II) presents estimates for a regression including interactions of the *Trouw* scores with the coverage rate of regional newspapers in each province that mention *Trouw* (yearly circulation per inhabitant). As can be seen from the table, none of these interactions are significant; the *F*-tests for the joint significance of the interaction terms are also rejected in both columns. The size of the interaction terms is typically small. A well-performing school in a market with a *Trouw* coverage rate one standard deviation above the mean gets an additional 0.16 students compared with such a school in a news market of average saturation. This effect is small, especially when one considers that any form of potential endogeneity of news coverage by regional newspapers would yield an upward bias. We have performed additional analyses using the content of the regional newspaper articles: one regression including interactions with the coverage rate when well-performing schools are mentioned, and another including interactions with the coverage rate when badly-performing schools are mentioned. No significant or substantial coefficient estimates were obtained. We thus conclude that direct exposure to the school-quality information cannot explain the effects found in Table 2.

The dissemination of information could alternatively be driven by market factors. In this case, either supply-side or demand-side factors could play a role. Dutch schools are financed based on the number of students they attract, so they have an incentive to disclose information that makes them more attractive and their competitors less attractive to potential customers. Such disclosure mechanisms are explained in models by, for example, Grossman and Hart (1980) and Jovanovic (1982). These authors argue that firms follow a reservation strategy in disclosing information about their quality. As secondary school choice is both one-off and considered to be an important decision, we also expect parents to search actively for information regarding the quality of schools. Rather than waiting until a newspaper boy delivers them some information about their choice set, it is likely that at least a subset of parents gathers the relevant information themselves. The demand for information about quality is higher the larger the related benefits are considered. The fact that we find the strongest effect of the *Trouw* scores

TABLE 6. Model extensions: regression coefficients from school-track, fixed effect regressions (dependent variable: first-year students in school track).

Interactions with	(I) Coverage rate, <i>Trouw</i> newspaper b/se	(II) Coverage rate, regional newspapers mentioning <i>Trouw</i> b/se	(III) Number of schools within 20 km b/se
Quality score at time <i>t</i>			
Not available	−0.791 (0.93)	−0.811 (0.89)	−0.749 (0.89)
Most negative	−4.940* (2.18)	−4.215** (1.55)	−4.047* (1.82)
Negative	−3.091*** (0.54)	−3.008*** (0.52)	−2.955*** (0.55)
Neutral	Reference	Reference	Reference
Positive	1.951*** (0.52)	1.907*** (0.53)	2.099*** (0.54)
Most positive	6.131* (2.45)	6.710* (2.71)	6.655* (2.60)
(Standardized) interaction variable with			
Not available	−0.427 (0.86)	−0.004 (0.42)	−1.312 (0.79)
Most negative	1.62 (2.27)	−1.109 (1.65)	0.301 (1.17)
Negative	0.139 (0.52)	0.155 (0.50)	−0.090 (0.49)
Neutral	Reference	Reference	Reference
Positive	0.158 (0.44)	−0.262 (0.41)	1.326* (0.60)
Most positive	−2.435 (2.95)	−2.580* (1.31)	1.785 (2.62)
<i>F</i> -stat scores	11.1	11.46	10.45
<i>p</i> -value	0.00	0.00	0.00
<i>F</i> -stat interactions	0.38	0.98	1.92
<i>p</i> -value	0.86	0.43	0.09
<i>N</i>	12,339	12,339	12,339

Notes: All additional controls (as in Table 2) included.
Standard errors between parentheses, clustered at the school level.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

for the most academic school track possibly indicates that parents of the brightest kids search most actively for information about quality.

Although our empirical design cannot distinguish between the supply- and demand-side stories, one would always expect larger information effects in more competitive markets. Column (III) of Table 6 displays evidence in this direction. We have estimated a regression including interactions of the *Trouw* scores with the normalized number of competitors in the area (i.e., schools within 20 km of each school). The estimates for these interaction terms are generally of the expected signs (such that the more competitors, the stronger the information effects) and the effect

for well-performing schools is significant.²⁵ The size of the interaction coefficients is, however, not very large. This lends partial credence to the idea that market forces are at play: well-performing schools may be promoting themselves, while informed consumers may be moving to better schools in more competitive markets.

5. Individual-Level Data Analysis

5.1. Individual-Level Data

So far, the main results in this paper are derived from school-level data. Since school choice is an individual decision, this section enriches our analysis with some results from an individual-level, cross-sectional dataset of one-off school choices. With these data, we cannot infer causal relationships, which hampers any direct interpretation of the coefficients describing the effects of the *Trouw* scores. The individual-level analysis does, however, address two issues that cannot be understood from the school-track-level estimation alone. First, we can assess the relative importance of schools' quality scores compared to distances from home. Second, the individual data allow us to estimate differences in the responses to quality scores of different socio-economic groups.

For this complementary analysis, we use an administrative dataset comprising one-third of the cohort of Dutch students that entered the first year of secondary school in September 2003. Using social-security numbers, we merged administrative records from Statistics Netherlands on the student level with administrative tax records on the level of the students' households. For all first-year students in the dataset, we constructed a relevant choice set of schools in their neighborhood. It is assumed that parents limited their school-searching behavior to only one school track, such that the school track in which a student was observed defines the choice set. For 26% of the children in our sample, we know this school track immediately from the 2003 administrative records, because they either attended single-track schools or were admitted to single-track classes. We use administrative records from the academic year 2005/2006 to retrieve the school tracks of the other students for whom tracks could not be recovered directly; in the third year of secondary school, the vast majority of students have transitioned from mixed-track to single-track classes. For about 30% of the students, however, we still fail to retrieve their school track through this procedure, as these students' schools do not register the school tracks of their students. Schools are not obliged to do this before the final year in which all students take centralized exams. These students are therefore left out of the analysis.

Table A.5 in the Online Appendix shows descriptive statistics of the individual-level data. We have a sample of 23,923 first-year students for whom we know household income and who chose to attend a school less than 20 km from home. Of these

25. The coefficient value of the effect of competition is stronger when one divides *Trouw*'s quality score into three categories (negative, neutral, and positive) instead of five.

students, 7,430 students attended the most academic track, 7,176 students attended the middle general track, and 9,317 students attended the lowest general track. On average, students had 29 school-track options within 20 km of their home, resulting in 670,272 possible combinations of individuals and school tracks. The average distance to travel to get to their nearest school ranges between 2.3 and 2.9 km for the different school tracks, while the average distance to the school actually chosen ranges between 3.9 and 4.3 km.

5.2. *Individual-Level Analysis*

Using these data, we estimate conditional logit regressions on the set of schools that children could have chosen. We define this set to consist of all schools that offered a particular school track and that lay within 20 km of the child’s home address.²⁶ In these regressions, we estimate the probability of choosing a certain school from one’s choice set, based on the characteristics of that school. The key control variable is the distance from a student’s home to each school. The set of other independent variables includes seven dummy variables for seven different categories of the relative, rank-ordered distance of each school from home and several school characteristics recorded at time $t - 3$, such as the total number of students that attended the school, the total number of first-year students, the number of school locations operated by the school management, the percentage of students in each school track, and the percentage of children from cultural minorities. We use standard maximum likelihood to estimate the parameters in the model.

The estimated odds ratios that are shown in Table 7 are for the full sample (Column (I)), for the most academic school track (Column (II)), for the middle academic school track (Column (III)), and for the lowest general school track (Column (IV)). The first row in Table 7 shows for each of the regressions the average, unconditional probability of a school being chosen. These base probabilities—ranging from 3% to 4%—are useful for interpreting the size of the odds ratios below. In all four regressions, the distance to a school in kilometers and the relative, rank-ordered distance of a school are clearly the most important determinants of school choice. These two factors explain between 89% (for academic-track students) and 98% (for lower-track students) of the 46% variation in choice behavior that is explained by observed characteristics. In the first column, the probability of a school being chosen goes down by 31% for each kilometer farther away from the child’s home address that it is located.²⁷

Table 7 also shows that the school-quality indicators do matter for individuals’ school choices, particularly when analyzing all school tracks together. According to the estimates, a school that has been ranked most negatively (“–”) is 27% less likely to be chosen compared to an identical school that received a neutral quality score, whereas a school that has been ranked moderately negatively (“–”) is 12% less likely

26. The results do not change when this radius is altered to 5, 10, or 15 km.
27. Our preferred specification of the distance variable is linear, as nonlinear specifications did not add much explanatory power.

TABLE 7. Odds ratios from conditional logit regressions in the individual-level analysis.

Dependent variable	(I) School choice within 20 km All first-year students	(II) School choice within 20 km Students in most academic track	(III) School choice within 20 km Students in middle academic track	(IV) School choice within 20 km Students in lowest general track
Selection				
<i>Average unconditional probability of choosing a school</i>				
	0.035	0.038	0.039	0.032
<i>Distance from home address to school, in kilometers</i>				
	0.685*** (0.005)	0.678*** (0.009)	0.666*** (0.010)	0.696*** (0.008)
<i>Relative, rank-ordered distance from home address to school</i>				
1st (closest school)	Reference	Reference	Reference	Reference
2nd to 3rd	0.846*** (0.019)	0.833*** (0.035)	0.811*** (0.035)	0.875*** (0.030)
4th to 6th	0.634*** (0.029)	0.629*** (0.052)	0.628*** (0.055)	0.632*** (0.046)
7th to 10th	0.463*** (0.042)	0.480*** (0.074)	0.484*** (0.082)	0.444*** (0.065)
11th to 15th	0.382*** (0.058)	0.383*** (0.105)	0.481*** (0.110)	0.342*** (0.091)
16th to 21st	0.393*** (0.074)	0.443*** (0.135)	0.457*** (0.145)	0.351*** (0.114)
22nd and beyond	0.472*** (0.084)	0.551*** (0.156)	0.692* (0.160)	0.370*** (0.127)
<i>Quality score published before t</i>				
Not available	0.700*** (0.032)	0.998 (0.062)	1.125 (0.061)	0.651*** (0.052)
Most negative	0.727** (0.123)	N.A. [#]	0.586* (0.217)	0.719* (0.154)
Negative	0.880*** (0.025)	0.995 (0.047)	0.934 (0.044)	0.854*** (0.039)
Neutral	Reference	Reference	Reference	Reference
Positive	1.088*** (0.022)	1.130** (0.040)	1.057 (0.040)	1.049 (0.038)
Most positive	1.082 (0.082)	0.940 (0.108)	1.186 (0.360)	0.937 (0.145)
N	670,272	194,834	184,655	290,783
R- squared	0.46	0.46	0.47	0.46

Notes: [#] In 2003, no VWO-schools were classified as performing in this category. Additional controls include: the total number of students that attended the school at $t - 3$, the total number of first-year students at $t - 3$, the number of school locations the school management operated at $t - 3$, and the percentage of students in each school track at $t - 3$. Standard errors of coefficients in parentheses and * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

to be chosen. Good quality scores increase the likelihood that a school track is chosen, with schools scoring well (“+”) being 9% more likely to be chosen. We do not find a significant effect for the most positive quality score (“++”), although an odds ratio larger than one is consistent with more students choosing to attend a school. Regarding the separate school-track samples (Columns (II)–(IV)), no clear pattern emerges from the results of our estimation, with most effects statistically insignificant.

Using the coefficients from the conditional logit estimations of the quality-score dummies and the coefficient of the distance variable, we can determine the implied ‘willingness-to-travel’ to a school of a certain quality. This measures the relative importance of the quality score compared to the importance of distance. The delta method is used to determine the standard errors of this willingness to travel. Table A.6 in the Online Appendix presents the implied willingness-to-travel to schools with certain quality scores. Note that each willingness-to-travel should be interpreted *ceteris paribus*, conditional on the relative, rank-ordered distance of a school. The estimates represent the willingness-to-travel relative to an average school, which is positive for negative quality scores, and reverse. Given that the average distance in our sample to students’ chosen schools is about four kilometers, the travelling distance clearly outweighs school-quality scores in each individual decision. Combining all school tracks, the largest (and statistically significant) distance students are willing to travel is 844 meters farther, so as to avoid a school assigned the most negative quality score by *Trouw*. To avoid a school that was rated “–” by *Trouw*, students would travel 339 meters. The distance that students are willing to travel to attend a school assigned a positive quality score by *Trouw* is only 222 meters.²⁸

One of our key findings in Section 4 is that the responses to quality information in school-choice decisions are mostly confined to the most academic education track. Since students in this track are, on average, from wealthier families, this raises questions about the importance of differences between the socio-economic situations of households for explaining differences between the tracks in terms of students’ responses to information about school quality. In other papers concerning quality information and school choice (e.g., Hastings, Kane, and Staiger 2005; Waslander, Pater, and Van der Weide 2010), it is implied that higher socio-economic groups are more responsive to school-quality information than other groups, thereby possibly increasing inequality in the quality of education enjoyed. We therefore re-estimate the baseline, conditional logit regressions with various interaction terms for the *Trouw* quality scores and individual characteristics. Such characteristics include income group, ethnicity, and main income source. The estimated coefficients of the interaction terms with a negative quality score (“–”) and with a positive quality score (“+”) can be found in Table A.7 in the appendix.

As Table A.7 shows, we find only two out of forty interaction coefficients to be significantly different from zero. We thus conclude that if parents of students in the academic track can be characterized by a stronger appreciation for the academic quality of schools and/or by lower search costs for information on quality, these characteristics are not correlated with income and ethnicity. This contrasts to the findings of Hastings et al. (2005), who do find substantial differences in responsiveness to information between socio-economic groups. At the same time, the second argument presented for

28. In order to assess the value of this number, we compare it to another school-track characteristic that students and parents care about: the percentage of children from cultural minorities. It is estimated that students are also willing to travel 222 meters farther to attend a school with only 0.3% immigrant children instead of a school with the average percentage of 5.6%.

the large effect on student inflow at academic tracks—that it is rational behavior to respond more to these scores for this particular track—gains strength because of these findings.

6. Conclusion

The picture that emerges is that parents do respond to the school-quality information that is published by *Trouw*. Information about quality thus affects school choice, even when the information is not actively and directly provided to consumers. Although our estimates suggest that the impact of the *Trouw* publication is generally small compared to the impact of the overall reputation of a school track or to the impact of school's distance from home, it is substantial for the subsample of academic school tracks. In particular, the inflow of first-year students goes up for this subsample by more than 20% (or by 16 to 18 students) after the track has received the most positive score (“++”) from *Trouw*. This impact is comparable to the effect found by Hastings et al. (2008).

The noisiness and/or volatility of the quality rankings may explain the small overall impact of the *Trouw* publication. If students and parents know that the score is largely transitory, they may attach relatively little value to it. We find some support for this hypothesis, as the track for which the *Trouw* score is most informative about and predictive of future quality-score outcomes happens to be the academic track. We also conjecture that knowledge of the *Trouw* scores may be low in general, but that knowledge of these scores may be higher among parents who actively search for the relevant information and among parents who are informed of the scores by well-performing schools. It may well be that such parents are more likely to send their children to academic school tracks.

There are reasons to believe that school-quality information is changing the educational market for the most academic track in the Netherlands. The recent oversubscription at excellent, academic-track schools indicates that the market for education is slowly approaching a new equilibrium. It remains to be seen what the implications of this equilibrium will be for secondary schools in the Netherlands. In the long run, quality could either converge or differentiation among schools could become stronger.

References

- Bishop, John H. (1998). “Privatizing Education: Lessons from Canada and Europe”. CAHRS Working Paper No. 98–21.
- Chay, Kenneth Y., Patrick J. McEwan, and Miguel Urquiola (2005). “The Central Role of Noise in Evaluating Interventions That Use Test Scores to Rank Schools.” *American Economic Review*, 95(4), 1237–1258.
- DellaVigna, Stefano and Ethan Kaplan (2007). “The Fox News Effect: Media Bias and Voting.” *Quarterly Journal of Economics*, 117, 1415–1451.

- Dijkstra, Anne B., Jaap Dronkers, and Sjoerd Karsten (2004). "Private Schools as Public Provision of Education: School Choice and Market Forces in the Netherlands." In *Educating Citizens. International Perspectives on Civic Values and School Choice*, edited by Patrick J. Wolf and Stephen Macedo. Brookings Institute Press, Washington, DC.
- Dijkstra, Anne B., Sjoerd Karsten, René Veenstra, and Adrie J. Visscher (2001). *Het Oog der Natie: Scholen op Rapport. Standaarden voor de Publicatie van Schoolprestaties*, Koninklijke van Gorcum, Assen.
- Downes, Thomas, A. and Jeffrey E. Zabel (2002). "The Impact of School Characteristics on House Prices: Chicago 1987–1991." *Journal of Urban Economics*, 52, 1–25.
- Dronkers, Jaap (1999). "Veranderden leerlingaantallen in het voortgezet onderwijs in het schooljaar 1998–1999 door de publicatie van inspectiegegevens en de berekening van het schoolcijfer door Trouw in oktober 1997?" *Tijdschrift voor Onderwijsresearch*, 24, 63–66.
- Enikolopov, Ruben, Maria Petrova, and Ekaterina Zhuravskaya (2011). "Media and Political Persuasion: Evidence from Russia." *American Economic Review*, 101(7), 3253–3285.
- Field, Simon, Małgorzata Kuczera, and Beatriz Point (2007). *No More Failures: Ten Steps to Equity in Education*. OECD, Paris.
- Figlio, David N. and Maurice E. Lucas (2004). "What's in a Grade? School Report Cards and the Housing Market." *The American Economic Review*, 94(3), 591–604.
- Gentzkow, Matthew A. (2006). "Television and Voter Turnout." *Quarterly Journal of Economics*, 121, 931–972.
- Grossman, Sanford J. and Oliver D. Hart (1980). "Disclosure Laws and Takeover Bids." *Journal of Finance*, 35, 323–334.
- Hastings, Justine S., Thomas J. Kane, and Douglas O. Staiger (2005). "Parental Preferences and School Competition: Evidence from a Public School Choice Program". NBER Working Paper No. 11805.
- Hastings, Justine S., Richard van Weelden, and Jeffrey Weinstein (2007). "Preferences, Information, and Parental Choice Behavior in Public School Choice." NBER Working Paper No. 12995.
- Hastings, Justine S. and Jeffrey Weinstein (2008). "Information, School Choice and Academic Achievement: Evidence from Two Experiments." *Quarterly Journal of Economics*, 123, 1373–1414.
- Jin, Ginger and Phillip Leslie (2003). "The Effect of Information on Product Quality: Evidence from Restaurant Hygiene Grade Cards." *Quarterly Journal of Economics*, 118, 409–451.
- Jin, Ginger and Phillip Leslie (2009). "Reputational Incentives for Restaurant Hygiene." *American Economic Journal: Microeconomics*, 1, 236–267.
- Jovanovic, Boyan (1982). "Truthful Disclosure of Information." *Bell Journal of Economics*, 13, 36–44.
- Kane, Thomas J., Stephanie K. Riegg, and Douglas O. Staiger (2006). "School Quality, Neighborhoods, and Housing Prices." *American Law and Economics Review*, 8, 183–212.
- Koning, Pierre and Karen van der Wiel (2012). "School Responsiveness to Quality Rankings: An Empirical Analysis of Secondary Education in The Netherlands." *De Economist*, 160, 339–355.
- Kling, Jeffrey R., Sendhil Mullainathan, Eldar Shafir, Lee Vermeulen, and Marian V. Wrobel (2008). *Misperception in Choosing Medicare Drug Plans*. Working paper, Harvard University.
- Pope, Devin G. (2009). "Reacting to Rankings: Evidence from 'America's Best Hospitals'." *Journal of Health Economics*, 28, 1154–1165.
- Waslander, Sietske, Cissy Pater, and Maartje van der Weide (2010). *Markets in Education: An Analytical Review of Empirical Research on Mechanisms in Education*. OECD Education Working Paper No. 52.
- Weeda, Frederiek (2010). "Vaker Loting voor Middelbare School." NRC Handelsblad, 30 January 2010.